Python Project Presentation

Python Bank Account Management System

A simple and interactive banking system built with Python to manage accounts, transactions and more...

Python NumPy jupyter Pickle



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Introduction & Objectives

About the Project

The Python Bank Account Management System is a Python-based project designed to simulate basic banking operations. It helps manage accounts, transactions and maintain records using Python, file handling and **NumPy**.



Project Objective

To apply fundamental Python programming concepts to a practical, real-world inspired banking scenario and understand how banking operations can be managed through a simple program.

Key Goals

- Manage bank accounts (create, view, update)
- Perform transactions (deposit, withdraw, transfer)
- Maintain transaction history
- Use file handling and NumPy for data storage and analysis

Technologies Used

Technologies Used

This project uses several Python libraries and tools to manage accounts, handle data and perform numerical operations efficiently.



Python

Core programming language used for building the entire application.

NumPy

Used for numerical computations and statistical analysis (e.g., transaction summary).

Pickle

Used for serialization and storage of Python objects (e.g., account data).

File Handling

Used to read and write data to files (e.g., transaction history).

Jupyter Notebook

Provides an interactive environment to write, run and test the code.

Features & Menu

Features & Menu

The system provides a simple, interactive and menu-driven interface to manage bank accounts and perform various banking operations.



Account Management
Create and manage bank accounts with unique numbers and account types.

Fund Transfer
Transfer money between accounts securely.

Deposit Money
Add funds to your account safely and easily.

Transaction History
View and track all your transactions.

Withdraw Money
Withdraw funds from your account with proper validation.

Summary & Reports
Get account balance, transaction summary and basic statistics (using NumPy).

Menu Options

1. Open a New Account
2. View Account Details
3. Deposit Money
4. Withdraw Money
5. Transfer Funds
6. Transaction History
7. Exit

Working & Implementation

Working & Implementation

The project follows a menu-driven approach where user inputs are processed through Python functions. Account data is stored using file handling and Pickle, and NumPy is used for numerical analysis.



bank_accounts.pkl
Stores account details using Pickle

transaction_history.txt
Stores transaction records in a text file

1. **User Input**
   User selects an option from the menu and provides the required input.
2. **Function Execution**
   The corresponding Python function processes the input (e.g., create account, deposit, withdraw, transfer).
3. **Data Storage**
   Account details are saved in bank_accounts.pkl using Pickle. Transaction records are written to transaction_history.txt.
4. **Numerical Analysis**
   NumPy is used to perform calculations such as total, average and other statistics from transaction data.
5. **Output Display**
   The result is shown to the user and updated data is stored for future use.

Working Flow

User selects menu option → Input is processed → Perform operation → Update files (Pickle & Text) → Display result

Implementation Details

Functions	Separate functions for each operation (create, deposit, withdraw, transfer, history, etc.).
Pickle	Used to serialize and store account objects in bank_accounts.pkl.
File Handling	Transaction details are saved in transaction_history.txt.
NumPy	Used for numerical computation (e.g., total and average transaction amounts) from transaction data.

Sample Output & Results

Project Output & Results

Here are some actual outputs from the Python Bank Account Management System, showing the account list, transaction history and summary results.

Account Holders & Balances

Account No.	Account Holders	Type	Balance (₹)
1001	Chetan Agrawal	Savings	6000.0
1002	Chetan Agrawal	Savings	10000.0
1003	Keshu	Savings	202000.0
1004	Rahul Goyal	Savings	59000.0

Transaction History (Account 1004)

```
--- Transaction History (Account 1004) ---
```

Date & Time	Type	Amount (₹)	Balance (₹)
2025-05-21 10:15:32	Opening Deposit	100000.00	100000.0
2025-05-22 14:30:17	Deposit	10000.00	110000.0
2025-05-23 16:45:09	Withdrawal	50000.00	60000.0
2025-05-24 11:20:33	Transfer Out	1000.00	59000.0

Sample Menu Output (Excerpt)

```
===== Bank Account Management System =====
1.  Open a New Account
2.  View Account Details
3.  Deposit Money
4.  Withdraw Money
5.  Transfer Funds
6.  Transaction History
7.  Summary & Reports
8.  Exit
Enter your choice: 6

--- Transaction History (Account 1004) ---
Date & Time          Type              Amount (₹)      Balance (₹)
2025-05-21 10:15:32  Opening Deposit    100000.00       100000.0
2025-05-22 14:30:17  Deposit            10000.00        110000.0
2025-05-23 16:45:09  Withdrawal         50000.00        60000.0
2025-05-24 11:20:33  Transfer Out       1000.00         59000.0
```

Transaction Summary (Account 1004)

Metric	Amount (₹)
Total Deposits / Incoming Transfers	110000.00
Total Withdrawals / Outgoing Transfers	51000.00
Average Transaction Amount	40250.00
Number of Transactions	4

Note

The above results are directly from the notebook's output. The account numbers, names and balances are as per your original data, with no changes.

Key Results

Total Accounts Created	4
Total Balance (All Accounts)	₹ 2,71,000
Total Transactions (Account 1004)	4
Average Transaction Amount (Account 1004)	₹ 40,250
System Status	Working Successfully

Conclusion & Future Scope

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This project successfully demonstrates a simple yet effective bank account management system using Python, with core features like account handling, transactions, file storage and data analysis.

Conclusion

Project Goal Achieved
The system meets all the core requirements — managing accounts, handling transactions, saving data and generating reports.

Efficient & Simple Implementation
Used Python with essential libraries like NumPy, Pickle and Jupyter Notebook for smooth and organized development.

Real-World Relevance
Helps in understanding how banking systems work and how data can be managed programmatically.

Data Insights
Provides basic analysis of transaction behavior using NumPy, which can be extended further.



Future Scope

User Authentication
Add login and password system for secure access to accounts.

Web / Mobile Interface
Develop a user-friendly UI using frameworks like Flask or Django.

Database Integration
Replace Pickle files with a real database (e.g., SQLite, MySQL) for better data management and security.

Advanced Features
Add features like interest calculation, loan management, fund transfer limits, and notifications.

Enhanced Data Analysis
Use more NumPy and data visualization (e.g., Matplotlib) for detailed reports and spending patterns.